Digital Financial Inclusion Predictor

Predicting Savings-Product Adoption

Your Name

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DIGITAL FINANCIAL INCLUSION PREDICTOR

Predicting Savings-Product Adoption

A Beginner Data Science Portfolio Project



Prepared by: Your Name

Programming language: R

Tools: Tidyverse, caret, pROC, Quarto

Responsible Data Science for Financial Inclusion

October 10, 2026

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1 1. Project Overview

1.1 1.1 Project Purpose

This project uses synthetic customer data to predict whether a customer is likely to adopt a savings product. It demonstrates how data science can support digital financial inclusion through data cleaning, exploratory analysis, predictive modelling, and responsible interpretation.

The model is intended to support customer education and voluntary outreach. It should not be used to automatically approve or deny loans or financial services.

1.2 1.2 Research Question

The main research question is:

> Which customer characteristics and financial behaviours are associated with savings-product adoption?

1.3 1.3 Project Objectives

The objectives are to generate and clean a synthetic dataset, explore adoption patterns, build a logistic regression model, evaluate model performance, interpret the results, and discuss business and ethical implications.

2 2. Business Problem

Financial institutions may have limited resources for reaching customers who could benefit from savings products. Customers differ in their mobile-money activity, income, savings behaviour, repayment history, access to agents, and exposure to financial education.

Understanding these patterns can help institutions design more relevant services and provide better support to underserved customers. The model should support human decision-making rather than replace it.

3 3. Dataset Description

The dataset contains approximately 1,000 synthetic customer records. It includes demographics, age group, location type, business ownership, monthly income, monthly expenses, mobile-money transactions, average transaction value, monthly savings, loan repayment rate, financial education, agent distance, application login frequency, customer-service contacts, and savings-product adoption.

The target variable is `0` for customers who did not adopt the product and `1` for customers who adopted it. The data is synthetic and contains no real names, phone numbers, identification numbers, account numbers, or exact addresses.

4 4. Data Generation and Cleaning

Synthetic customer records were generated in R to represent realistic digital-finance behaviours. Random variation was included so that the target outcome was not perfectly predictable.

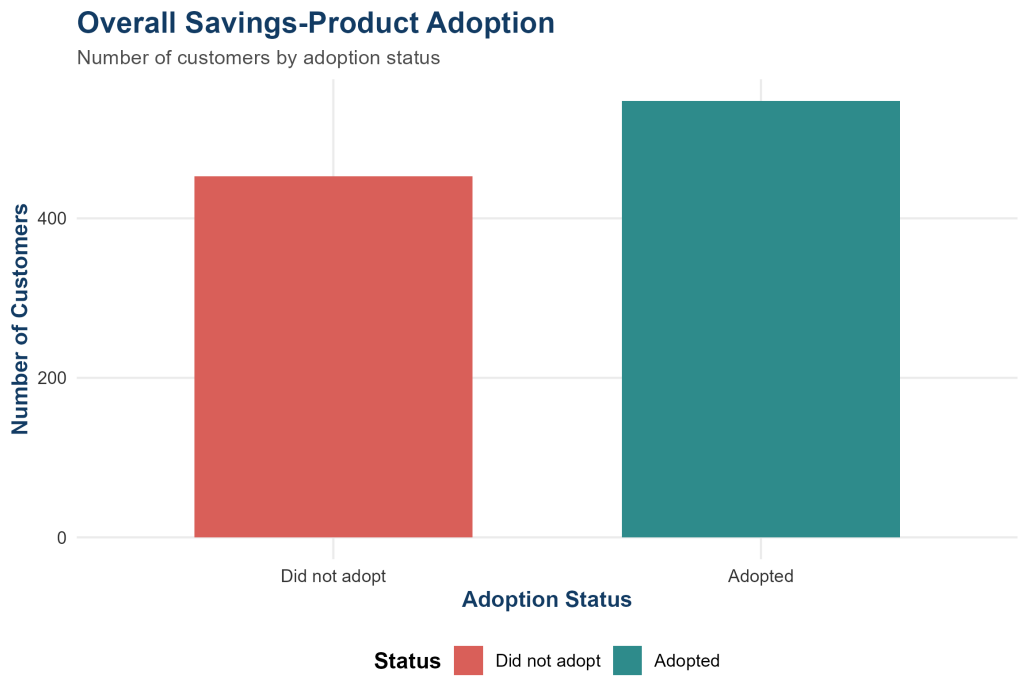
The data was checked for missing values, duplicate records, duplicate customer IDs, invalid financial values, incorrect data types, and inconsistent categorical values. The cleaning process included removing duplicates, standardising categories, handling missing values, checking numeric ranges, and validating the target variable.

The cleaned dataset was saved as `data/processed/customers_clean.csv` while the raw dataset remained unchanged.

5 5. Exploratory Data Analysis

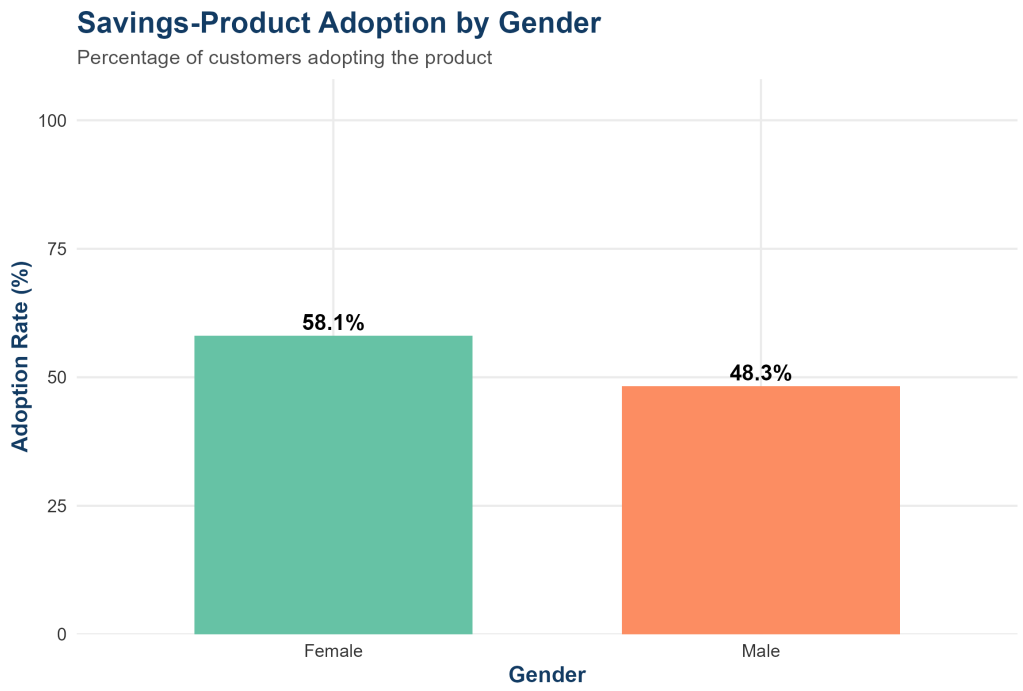
The following visualisations examine customer adoption patterns and financial behaviour.

5.1 5.1 Overall Savings-Product Adoption



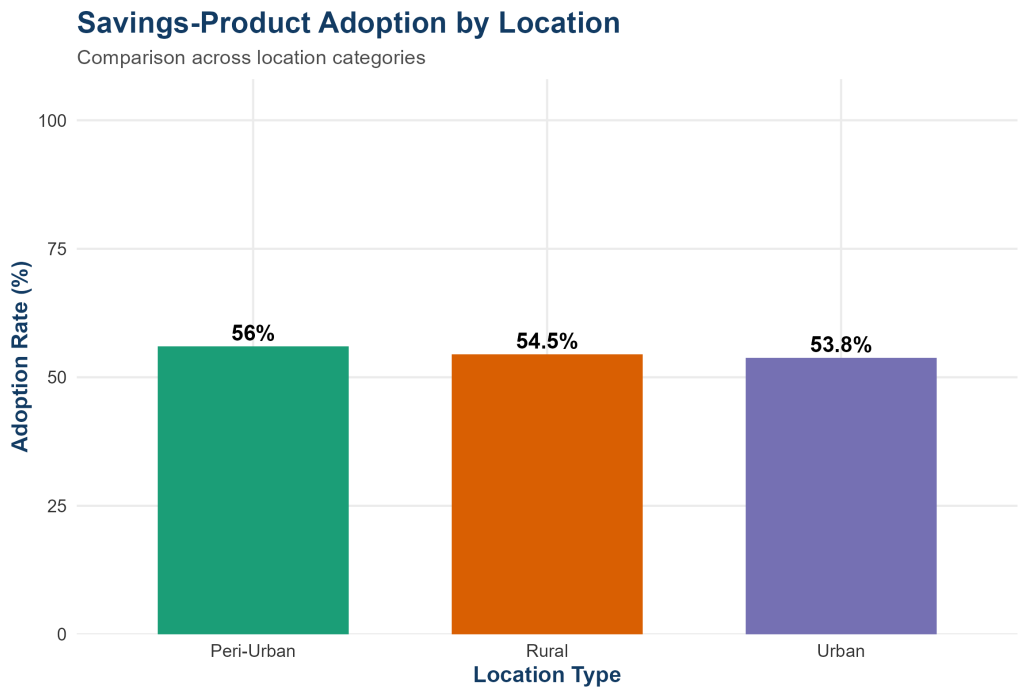
This chart shows the number of customers who adopted and did not adopt the savings product.

5.2 5.2 Adoption by Gender



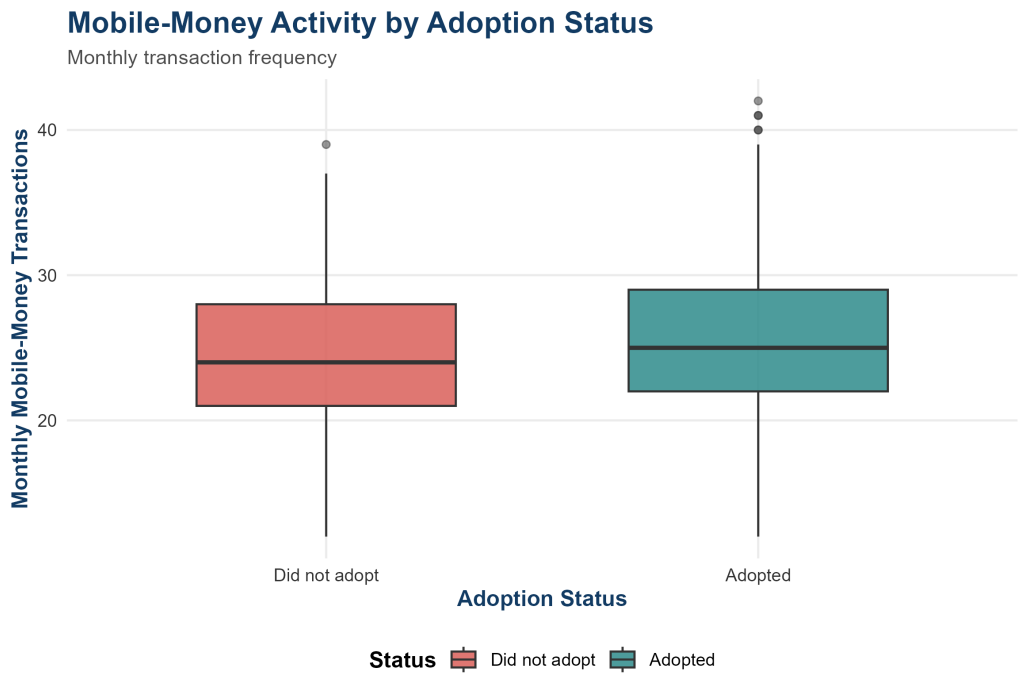
This chart compares adoption rates between gender groups. Differences may help identify where additional financial education or outreach could be useful.

5.3 5.3 Adoption by Location



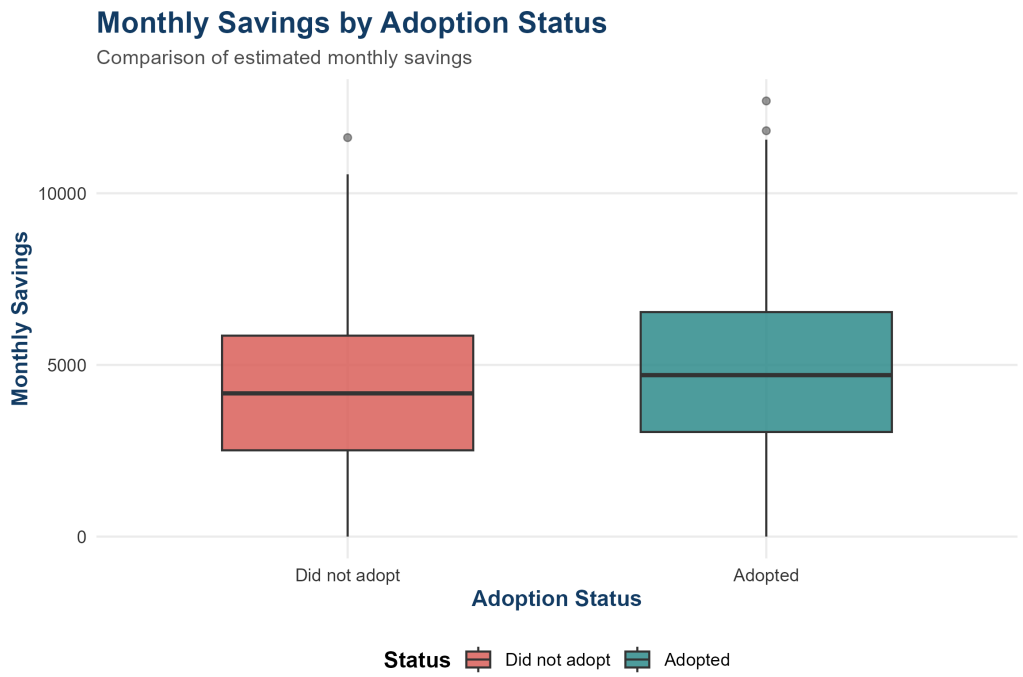
This chart compares adoption across urban, peri-urban, and rural customers. Differences may reflect variation in access, connectivity, agent distance, or financial awareness.

5.4 5.4 Mobile-Money Activity



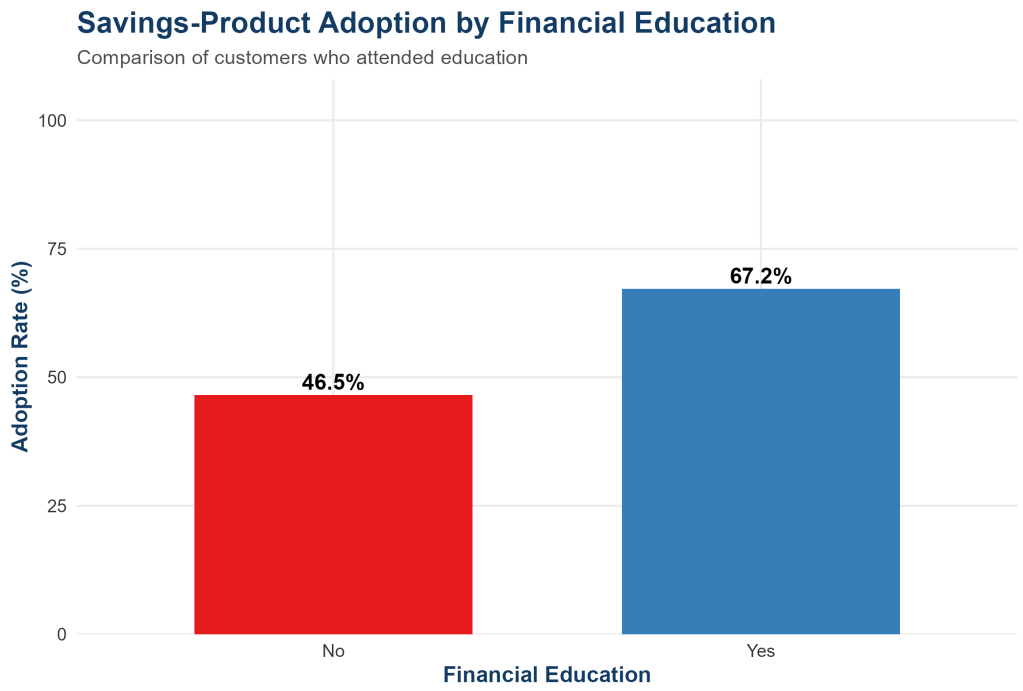
This chart compares monthly mobile-money activity between adopters and non-adopters.

5.5 5.5 Monthly Savings



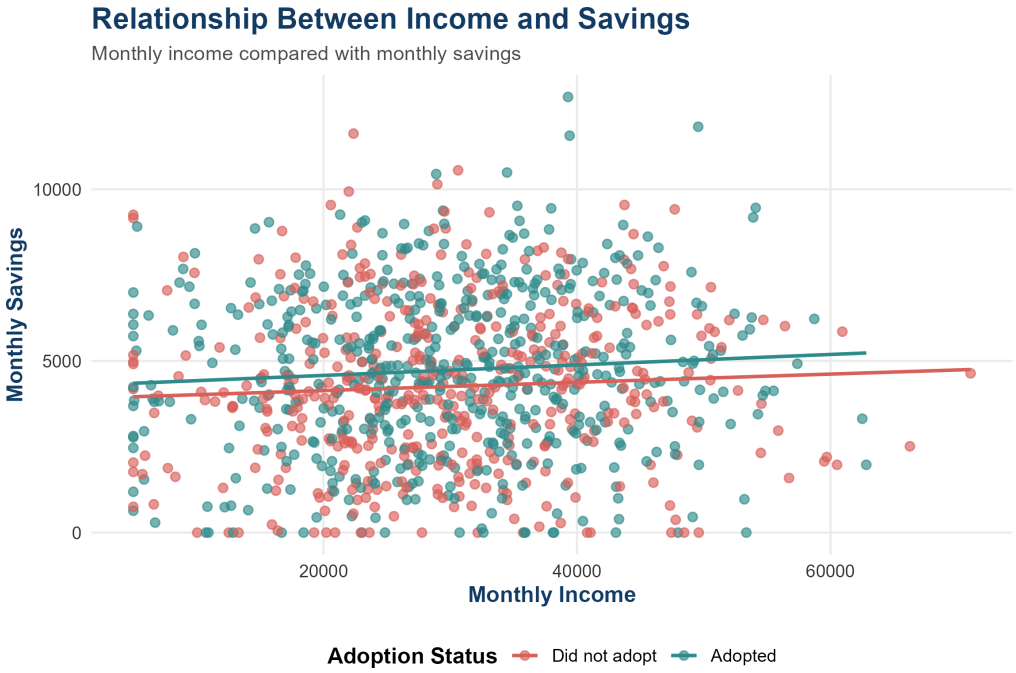
This chart compares monthly savings levels between adopters and non-adopters.

5.6 5.6 Financial Education



This chart examines whether customers who attended financial education showed different adoption rates.

5.7 5.7 Income and Savings



This scatter plot examines the relationship between monthly income and monthly savings.

6 6. Predictive Modelling

6.1 6.1 Model Selection

A logistic regression model was selected as the baseline model because it is relatively easy to interpret and explain to business stakeholders. It produces probabilities between zero and one that can be converted into adoption classifications.

6.2 6.2 Training and Testing Data

The data was divided into 80% training data and 20% testing data. The training data was used to estimate model parameters, while the testing data was used to evaluate performance on records not used during training.

6.3 6.3 Model Variables

The model used demographic, financial, and digital-engagement variables, including gender, age group, location, business ownership, income, expenses, mobile-money transactions, transaction value, savings, repayment rate, financial education, agent distance, application logins, and customer-service contacts.

The trained model was saved as `models/logistic_regression_model.rds`.

7 7. Model Performance

7.1 7.1 Performance Metrics

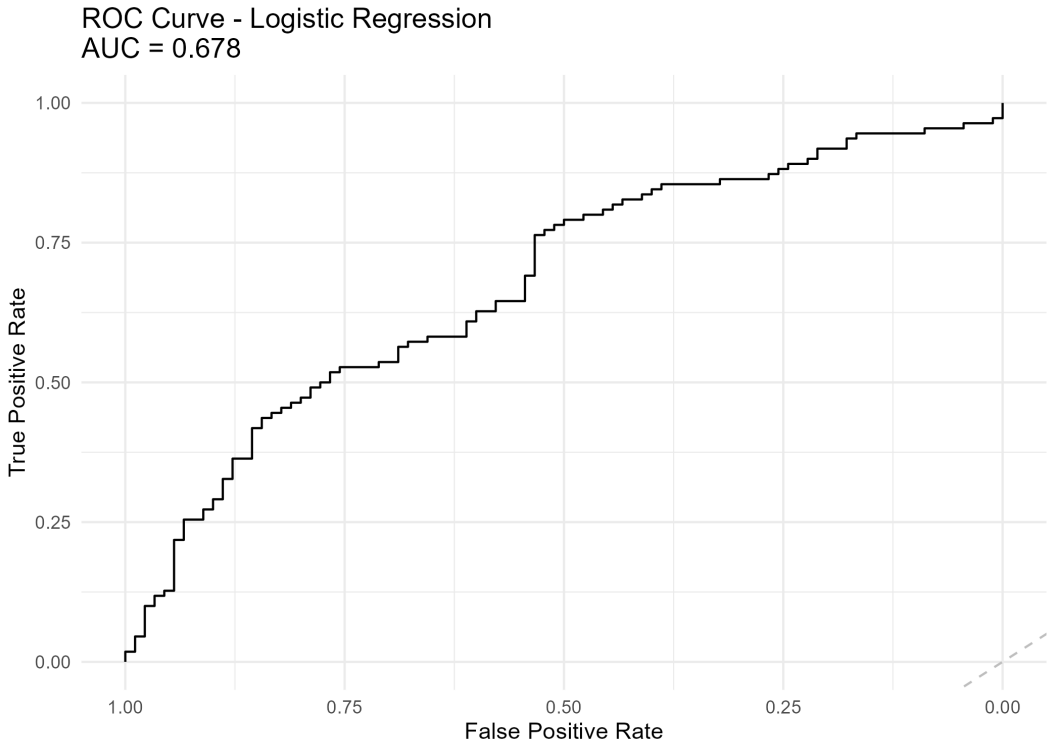
The model-performance results are shown below.

Table 7.1: Logistic regression model performance

Metric	Result
Accuracy	0.660
Sensitivity/Recall	0.764
Specificity	0.533
Precision	0.667
F1-score	0.712
ROC-AUC	0.678

The model achieved an ROC-AUC of approximately **0.678**. This indicates weak-to-moderate ability to distinguish between adopters and non-adopters. The model performs better than random guessing, represented by an ROC-AUC of 0.500, but there is room for improvement.

7.2 7.2 ROC Curve



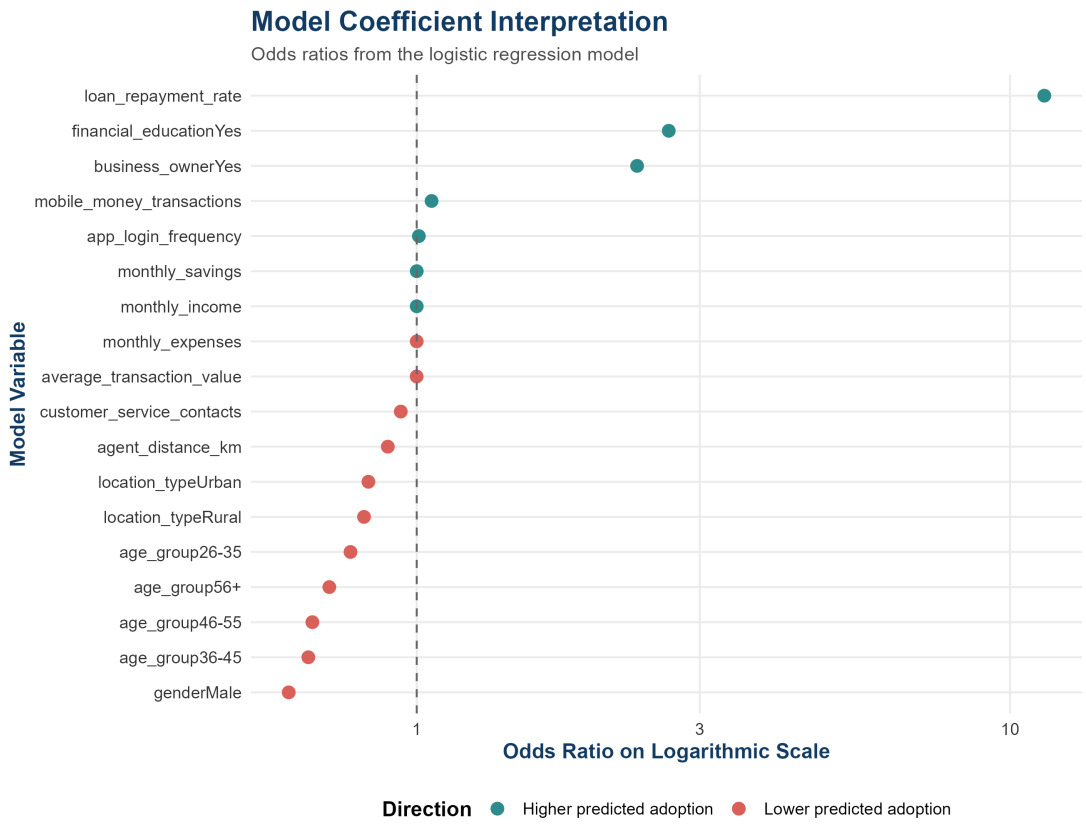
The ROC curve shows how well the model separates adopters from non-adopters across different classification thresholds.

7.3 7.3 Metric Interpretation

Accuracy measures the proportion of correct predictions. Sensitivity or recall measures the proportion of actual adopters correctly identified. Specificity measures the proportion of non-adopters correctly identified. Precision measures the proportion of customers classified as adopters who actually adopted. The F1-score balances precision and recall.

8 8. Model Interpretation

8.1 8.1 Coefficient Plot



The coefficient plot summarises the direction and strength of relationships estimated by the logistic regression model.

8.2 8.2 Important Predictors

The model showed positive associations between adoption and loan repayment rate, financial education, and business ownership. Agent distance had a negative association with predicted

adoption, which may suggest that customers farther from financial-service agents face greater access barriers.

These results are associations from synthetic data. They do not prove that the variables directly cause savings-product adoption.

8.3 8.3 Coefficients and Odds Ratios

Table 8.1: Selected model coefficients and odds ratios

Variable	Estimate	Odds Ratio	P-value
loan_repayment_rate	2.436	11.424	0.0001
financial_educationYes	0.978	2.660	0.0000
business_ownerYes	0.855	2.352	0.0000
genderMale	-0.497	0.609	0.0027
age_group36-45	-0.421	0.657	0.0742
age_group46-55	-0.405	0.667	0.1389
age_group56+	-0.339	0.712	0.3933
age_group26-35	-0.257	0.773	0.2668
location_typeRural	-0.205	0.815	0.2968
location_typeUrban	-0.187	0.829	0.3340

An odds ratio greater than 1 indicates higher predicted odds of adoption, while an odds ratio below 1 indicates lower predicted odds. These results should be interpreted as associations rather than proof of causation.

9 9. Findings

The logistic regression model achieved an ROC-AUC of approximately **0.678**, indicating weak-to-moderate ability to distinguish between customers who adopted the savings product and those who did not.

The model results showed positive associations between adoption and loan repayment rate, financial education, and business ownership. Agent distance had a negative association with predicted adoption.

The reported model-performance values included specificity of **0.533**, precision of **0.667**, F1-score of **0.712**, and ROC-AUC of **0.678**. The complete performance table appears in Section 7.

These results are based on synthetic data and do not prove that the variables directly cause savings-product adoption.

10 10. Business Recommendations

10.1 10.1 Improve Financial Education

Financial institutions could provide accessible financial-education materials through agents, mobile applications, SMS messages, and community programmes.

10.2 10.2 Support Appropriate Outreach

The model could help identify customers who may benefit from information about savings products. Any outreach should be voluntary, transparent, and relevant to customer needs.

10.3 10.3 Improve Access to Agents

The negative association between agent distance and adoption suggests that institutions could consider agent locations when planning customer-support activities. Digital alternatives may also help customers who are far from physical service points.

10.4 10.4 Support Small-Business Customers

Business owners may have particular needs related to cash flow, savings, working capital, and seasonal income. Institutions could design flexible savings products and financial-education programmes for these customers.

10.5 10.5 Use Human Review

The model should support human decision-making rather than replace it. Financial officers should consider customer preferences, explanations, and circumstances before making recommendations.

11 11. Ethical Considerations

A real-world implementation would require informed customer consent, data minimisation, secure storage, encryption, controlled access, fairness testing, human review, transparent explanations, and continuous monitoring.

The model should not automatically approve or deny financial services. It should not be used to exclude customers from financial opportunities based only on an automated prediction.

12 12. Limitations

The dataset is synthetic and does not represent actual customers. The findings may differ from results obtained using real-world data. The model identifies statistical associations, but associations do not prove causation.

The model would require further testing on representative real-world data, data from different periods, fairness across customer groups, model calibration, monitoring for drift, and review by financial and compliance professionals.

13 13. Future Improvements

Future improvements could include comparing logistic regression with random forest, testing different probability thresholds, evaluating fairness across gender and location groups, using a larger representative dataset, adding customer-feedback variables, building a Shiny application, adding model monitoring, and using cross-validation.

14 14. Conclusion

This project demonstrates how data science can support digital financial inclusion by identifying patterns associated with savings-product adoption.

The results suggest that financial education, digital activity, loan repayment behaviour, business ownership, and access to financial agents may be useful factors for understanding customer adoption.

Responsible use of predictive analytics could help financial institutions provide better education and more relevant services while protecting customer privacy and avoiding unfair decisions.

The model should be treated as a decision-support tool and not as a replacement for human judgement.

DIGITAL FINANCIAL INCLUSION PREDICTOR

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